01

Quest —
AI Support Agent

Support cases → grounded AI recommendations, written straight back into Dynamics 365 / ServiceNow

BUSINESS VALUE

~30–60s case → recommendation, 24/7 hands-off

INDUSTRIES

Cloud Infra · ITSM · Enterprise Support

BUILD STATUS

MVP deployed · 4-phase rollout

PRINCIPLE

AI proposes; it only acts behind confidence gates

PROBLEM

Engineers triage every case by hand — routing, root-cause hunting and fix research are slow, inconsistent, and repeat work already solved in past tickets.

OBJECTIVE

Route, diagnose and recommend a fix on every case in ~30–60s — grounded in real case history — so engineers act instead of investigate.

SOLUTION

A stateless orchestrator runs a Routing → Diagnosis → Recommendation → Mitigation pipeline on each case, posts one grounded note back, and safely auto-remediates only low-risk, high-confidence issues — every decision audited.

SYSTEMS INTEGRATED

Dynamics 365 | **ServiceNow** | **Azure OpenAI** | **MCP Tools** | **Postgres + pgvector**

FEATURES

- Webhooks → dedupe → state machine
- Skill- & shift-aware routing
- Similarity-grounded diagnosis (pgvector)
- Hot Fix + Ultimate Fix, allowlisted links
- Guarded auto-remediation + kill switch

USE CASES

- K8s / Karpenter incidents auto-triaged
- Reopened cases → escalated recommendation
- Manager SLA-risk dashboard
- 👍/👎 feedback regenerates the answer
- MCP tools reused by Copilot Studio

BENEFITS

- ~30–60s case → recommendation, 24/7
- Grounded — no hallucinated fixes
- Graceful degradation, no single point of failure
- Safety-first: suggests, acts only behind gates
- Full audit trail of every AI decision

DELIVERY ROADMAP — 4 PHASES

Phase 1 LIVE
Orchestration & Grounded Recommendations

Now:
Routing → Diagnosis → Recommendation, pgvector, D365 write-back, audit log

Phase 2
MCP Engine & Tooling

Now: reasoning/knowledge/dataverse servers, dual-engine flag → Copilot Studio

Phase 3
Guarded Auto-Remediation

Now: safety gate + kill switch + verify-then-revert → real runbook execution

Phase 4
Feedback Loop & Manager Ops

Now: 👍/👎 regeneration, KB ranking, SLA dashboard → approve-and-execute flow